1. Sample Database

Table: Customers

customer_id	customer_name	city	customer_type
1	Rahul	Delhi	Premium
2	Meena	Mumbai	Regular
3	Arjun	Bangalore	Premium
4	Kavya	Delhi	Regular
5	Suresh	Chennai	Premium
6	Anjali	Mumbai	Regular
7	Vikram	Delhi	Regular
8	Pooja	Bangalore	Premium
9	Devi	Chennai	Regular

Table: Products

product_id	product_name	category	price
1	Laptop	Electronics	55000
2	Mobile	Electronics	25000
3	Chair	Furniture	7000
4	Table	Furniture	9000
5	Headphones	Electronics	3000
6	Bookshelf	Furniture	6000

Table: Orders

order_id	customer_id	product_id	quantity	order_date	amount
1	1	1	1	2023-01-05	55000

2	2	2	1	2023-01-10	25000
3	3	3	2	2023-02-01	14000
4	1	5	2	2023-02-15	6000
5	4	4	1	2023-03-03	9000
6	5	2	2	2023-03-20	50000
7	6	4	3	2023-04-02	18000
8	3	5	1	2023-04-15	3000
9	7	3	1	2023-05-01	7000
10	8	1	1	2023-05-20	54000
11	2	3	2	2023-06-02	12000
12	5	4	1	2023-06-15	8500

Note: "amount" is the total paid for that order (it may reflect quantity, discounts, etc., and does not need to equal price × quantity exactly).

2. Database Setup — CREATE TABLE & INSERT Statements

Run the script below in your SQL client (MySQL / SQL Server / Oracle / PostgreSQL — syntax may need minor tweaks) to build the exact database used throughout this worksheet.

Step 1: Create the tables

```
CREATE TABLE Customers (
  customer_id    INT PRIMARY KEY,
  customer_name  VARCHAR(50),
  city           VARCHAR(50),
  customer_type  VARCHAR(20)
);

CREATE TABLE Products (
  product_id     INT PRIMARY KEY,
  product_name   VARCHAR(50),
  category       VARCHAR(30),
  price          DECIMAL(10,2)
);

CREATE TABLE Orders (
  order_id       INT PRIMARY KEY,
  customer_id    INT REFERENCES Customers(customer_id),
  product_id     INT REFERENCES Products(product_id),
  quantity       INT,
  order_date     DATE,
  amount         DECIMAL(10,2)
);
```

Step 2: Insert Customers records

```
INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (1, 'Rahul', 'Delhi', 'Premium');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (2, 'Meena', 'Mumbai', 'Regular');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (3, 'Arjun', 'Bangalore', 'Premium');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (4, 'Kavya', 'Delhi', 'Regular');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (5, 'Suresh', 'Chennai', 'Premium');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (6, 'Anjali', 'Mumbai', 'Regular');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (7, 'Vikram', 'Delhi', 'Regular');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (8, 'Pooja', 'Bangalore', 'Premium');

INSERT INTO Customers (customer_id, customer_name, city, customer_type) VALUES (9, 'Devi', 'Chennai', 'Regular');
```

Step 3: Insert Products records

```
INSERT INTO Products (product_id, product_name, category, price) VALUES (1, 'Laptop', 'Electronics', 55000);

INSERT INTO Products (product_id, product_name, category, price) VALUES (2, 'Mobile', 'Electronics', 25000);

INSERT INTO Products (product_id, product_name, category, price) VALUES (3, 'Chair', 'Furniture', 7000);

INSERT INTO Products (product_id, product_name, category, price) VALUES (4, 'Table', 'Furniture', 9000);

INSERT INTO Products (product_id, product_name, category, price) VALUES (5, 'Headphones', 'Electronics', 3000);

INSERT INTO Products (product_id, product_name, category, price) VALUES (6, 'Bookshelf', 'Furniture', 6000);
```

Step 4: Insert Orders records

```
INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (1, 1, 1, 1, '2023-01-05', 55000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (2, 2, 2, 1, '2023-01-10', 25000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (3, 3, 3, 2, '2023-02-01', 14000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (4, 1, 5, 2, '2023-02-15', 6000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (5, 4, 4, 1, '2023-03-03', 9000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (6, 5, 2, 2, '2023-03-20', 50000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (7, 6, 4, 3, '2023-04-02', 18000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (8, 3, 5, 1, '2023-04-15', 3000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (9, 7, 3, 1, '2023-05-01', 7000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (10, 8, 1, 1, '2023-05-20', 54000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (11, 2, 3, 2, '2023-06-02', 12000);

INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date, amount) VALUES (12, 5, 4, 1, '2023-06-15', 8500);
```

3. Single-Row Subqueries

A single-row subquery returns exactly one row and one column, so it is used with a plain comparison operator: =, >, <, >=, <=, or <>.

Q1. Find the name and price of the most expensive product.

Answer:

```
SELECT product_name, price

FROM Products

WHERE price = (SELECT MAX(price) FROM Products);
```

Result: *Laptop | 55000 (1 row)*

Q2. Find the order (order_id, customer_id, amount) with the smallest order amount.

Answer:

```
SELECT order_id, customer_id, amount
FROM Orders
WHERE amount = (SELECT MIN(amount) FROM Orders);
```

Result: *order_id = 8, customer_id = 3, amount = 3000 (1 row)*

Q3. Find the name of the customer who placed order number 5

.Answer:

```
SELECT customer_name
FROM Customers
WHERE customer_id = (SELECT customer_id FROM Orders WHERE order_id = 5);
```

Result: *Kavya (1 row)*

Q4. Find the category of the product that was purchased in order number 1.

Answer:

```
SELECT category
FROM Products
WHERE product_id = (SELECT product_id FROM Orders WHERE order_id = 1);
```

Result: *Electronics (1 row)*

Q5. Find the name of the product involved in the order with the highest amount.

Answer:

```
SELECT product_name
FROM Products
WHERE product_id = (
  SELECT product_id FROM Orders WHERE amount = (SELECT MAX(amount) FROM Orders)
);
```

Result: *Laptop (1 row)*

4. Multi-Row Subqueries — IN

The IN operator checks whether a value matches any value in the list returned by the subquery. NOT IN checks that it matches none of them.

Q1. Find the names of customers who live in 'Delhi' or 'Mumbai'.

Answer:

```
SELECT customer_name
FROM Customers
WHERE city IN ('Delhi', 'Mumbai');
```

Result: *Rahul, Meena, Kavya, Anjali, Vikram (5 rows)*

Q2. Find the names of products that have been ordered at least once.

Answer:

```
SELECT product_name
FROM Products
WHERE product_id IN (SELECT product_id FROM Orders);
```

Result: *Laptop, Mobile, Chair, Table, Headphones (5 rows)*

Q3. Find the names of customers who have placed at least one order.

Answer:

```
SELECT customer_name
FROM Customers
WHERE customer_id IN (SELECT customer_id FROM Orders);
```

Result: *All customers except Devi (8 rows)*

Q4. Find products that have never been ordered (use NOT IN).

Q5. Find customers whose city matches a city where at least one 'Premium' customer lives.

5. Multi-Row Subqueries — ANY / SOME

ANY (and its synonym SOME) makes the outer comparison true if it holds for at least one value returned by the inner query. For example, "> ANY" means greater than the smallest value returned.

Q1. Find products priced greater than ANY product in the 'Furniture' category.

Answer:

```
SELECT product_name, price
FROM Products
WHERE price > ANY (SELECT price FROM Products WHERE category = 'Furniture');
```

Result: *Laptop, Mobile, Chair, Table (4 rows)*

Q2. Find orders with an amount greater than ANY order placed by 'Arjun'.

Answer:

```
SELECT order_id, amount
FROM Orders
WHERE amount > ANY (
  SELECT amount FROM Orders
  WHERE customer_id = (SELECT customer_id FROM Customers WHERE customer_name = 'Arjun')
);
```

Result: *All orders except order_id 8 (11 rows)*

Q3. Find products priced less than SOME product in the 'Electronics' category.

Answer:

```
SELECT product_name, price
FROM Products
WHERE price < SOME (SELECT price FROM Products WHERE category = 'Electronics');
```

Result: *Mobile, Chair, Table, Headphones, Bookshelf (5 rows)*

Q4. Find orders with an amount greater than ANY order placed by a 'Regular' customer.

Q5. Find orders with an amount less than ANY order placed by a customer from 'Mumbai'.

6. Multi-Row Subqueries — ALL

ALL makes the outer comparison true only if it holds against every value returned by the inner query. For example, "> ALL" means greater than the largest value returned, and "< ALL" means less than the smallest value returned.

Q1. Find products priced higher than ALL products in the 'Furniture' category.

Answer:

```
SELECT product_name, price
FROM Products
WHERE price > ALL (SELECT price FROM Products WHERE category = 'Furniture');
```

Result: *Laptop, Mobile (2 rows)*

Q2. Find orders with an amount greater than ALL orders placed by 'Meena'.

Answer:

```
SELECT order_id, amount
FROM Orders
WHERE amount > ALL (
  SELECT amount FROM Orders
  WHERE customer_id = (SELECT customer_id FROM Customers WHERE customer_name = 'Meena')
);
```

Result: *order_id 1, 6, 10 (3 rows)*

Q3. Find products priced lower than ALL products in the 'Furniture' category.

Answer:

```
SELECT product_name, price
FROM Products
WHERE price < ALL (SELECT price FROM Products WHERE category = 'Furniture');
```

Result: *Headphones (1 row)*

Q4. Find orders with an amount greater than ALL orders placed for 'Furniture' category products.

Q5. Find orders with an amount less than ALL orders placed for 'Furniture' category products.

7. Nested Subqueries

A nested subquery is a subquery placed inside another subquery. SQL evaluates from the innermost query outward, and each result feeds the query one level up.

Q1. Find the name(s) of the customer(s) who ordered the most expensive product.

Answer:

```
SELECT customer_name
FROM Customers
WHERE customer_id IN (
  SELECT customer_id FROM Orders
  WHERE product_id = (
          SELECT product_id FROM Products
          WHERE price = (SELECT MAX(price) FROM Products)
  )
);
```

Result: *Rahul, Pooja (2 rows — both bought the Laptop)*

Q2. Find the category of the product bought in the highest-amount order.

Answer:

```
SELECT category
FROM Products
WHERE product_id = (
  SELECT product_id FROM Orders
  WHERE amount = (SELECT MAX(amount) FROM Orders)
```

```
);
```

Result: *Electronics (1 row)*

Q3. Find the city (or cities) of the customer(s) who bought the cheapest product.

Answer:

```
SELECT city
FROM Customers
WHERE customer_id IN (
  SELECT customer_id FROM Orders
  WHERE product_id = (
            SELECT product_id FROM Products
            WHERE price = (SELECT MIN(price) FROM Products)
  )
);
```

Result: *Delhi, Bangalore (2 rows — Rahul and Arjun bought the Headphones)*

Q4. Find the names of customers who live in the same city as the customer who placed the highest-amount order.

Q5. Find the name(s) of products ordered by 'Rahul' that cost more than the average price of all products.

8. Correlated Subqueries

A correlated subquery references a column from the outer query, so it cannot run on its own — it is re-evaluated once for every row the outer query processes.

Q1. Find customers who have placed more than one order.

Answer:

```
SELECT customer_name
FROM Customers c
WHERE (SELECT COUNT(*) FROM Orders o WHERE o.customer_id = c.customer_id) > 1;
```

Result: *Rahul, Meena, Arjun, Suresh (4 rows)*

Q2. Find the highest-amount order placed by each customer.

Answer:

```
SELECT o1.order_id, o1.customer_id, o1.amount
FROM Orders o1
WHERE o1.amount = (
  SELECT MAX(o2.amount) FROM Orders o2
  WHERE o2.customer_id = o1.customer_id
);
```

Result: *order_id 1, 2, 3, 5, 6, 7, 9, 10 (8 rows — one top order per customer)*

Q3. Find products priced higher than the average price of products in their own category.

Answer:

```
SELECT p1.product_name, p1.category, p1.price
FROM Products p1
WHERE p1.price > (
  SELECT AVG(p2.price) FROM Products p2
  WHERE p2.category = p1.category
);
```

Result: *Laptop (Electronics), Table (Furniture) (2 rows)*

Q4. Find customers whose order amount is greater than the average order amount of all customers from their own city.

Q5. Find the second-lowest priced product within each category.

9. EXISTS and NOT EXISTS

EXISTS only checks whether the subquery returns any rows at all (true/false) — the actual values don't matter. NOT EXISTS is true when the subquery returns no rows.

Q1. Find customers who have placed at least one order (use EXISTS).

Answer:

```
SELECT customer_name
FROM Customers c
WHERE EXISTS (SELECT 1 FROM Orders o WHERE o.customer_id = c.customer_id);
```

Result: *All customers except Devi (8 rows)*

Q2. Find customers who have never placed an order (use NOT EXISTS).

Answer:

```
SELECT customer_name
FROM Customers c
WHERE NOT EXISTS (SELECT 1 FROM Orders o WHERE o.customer_id = c.customer_id);
```

Result: *Devi (1 row)*

Q3. Find products that have been ordered at least once (use EXISTS).

Answer:

```
SELECT product_name
FROM Products p
WHERE EXISTS (SELECT 1 FROM Orders o WHERE o.product_id = p.product_id);
```

Result: *Laptop, Mobile, Chair, Table, Headphones (5 rows)*

Q4. Find products that have never been ordered (use NOT EXISTS).

Q5. Find customers who have not ordered any product from the 'Electronics' category (use NOT EXISTS).